

Karl Friston's Free Energy Principle and Predictive Coding - The StrongFit Podcast Episode 020

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I All right guys we are back This is gonna be this is the first one back after only week where we didn't produce an episode Julien: that break. Yeah Tyler: Actually,...' Julien: I mean Dubai and other places Tyler: It was kind of funny though. I noticed that When we when we had started this whole thing like before it moved here before any of the thing I remember we've we kind of assumed we were gonna have to do this in like ten episode seasons just because of traveling, you know, what kind of thought it was gonna be tough to put together So the fact that we got 20 episodes Julien: Is that how many we did? Tyler: We did it 19 actually, it was 19 without a single Yeah, so 19 weeks 19 episodes. Julien: You know that people don't care All right. They gotta be like what are you Tyler:guys we're doing our best? Julien: Yes, like I'm in the car. I got nothing to do Tyler: Actually it was funny though this morning. I woke up to frantic messages from Julien of like Lists of studies and all these other things to go through and and I think Kyla was taking pictures of you with your whiteboard Julien: Yes. Well, there was some interesting. Yeah, yeah Tyler: Now one of my favorite things about this this happens I would say probably about once a month on average where you like you like get a wild hair up your ass and you're like Tyler dragging in the rabbit hole, first thing Tyler: Just like the the way there's Julien: No and there's no foreplay on this one Tyler: It's like when also when Julian says You got five minutes for a call. What that means is you have 90 minutes you call Julien: And it's gonna be interesting. Tyler: Yes, but But today you had messaged me a few things that were actually really really lined up with what we wanted to talk about today anyways Julien: What you do, which is Karl Friston stuff Tyler: Without going into all of what his stuff is because that's gonna be the bulk of this episode Julien: The concept.. Tyler: Yeah. Yeah, but the guy himself is a What is this? He was a neuro was it a neuroscientist or a neuroimaging scientist Julien: He is a neuroscientist He started as a psychiatrist. Okay, that was that's where he wanted to go, it was the root of mental illness It's schizophrenia stuff like that. So that's

what he wanted to do, to fix Whatever you want to call it to cure was mental illness and everything and he moved into Moneo science, he's an excellent mathematician. Tyler: As one of the things that I had seen I had some notes here from Dan article more on the guy than on his work, but Was it like? Basically his his neuroscience stuff. Now that he had done is all of his uh brain imaging and other stuff is actually like his kind of like his hobby now while he's driving really really really far forward now unto his Yeah free energy stuff. Yeah Julien: Well, yeah, and then he also is driving the field of artificial intelligence. Yeah, which I'm not I don't even know that the people in the neuroscience Understand the the leaps forward made in the field of a artificial intelligence because of him Like all the new stuff that we see coming up is a direct follow-up have on his work. That's why like it's not artificial intelligence anymore. It's basically artificial learning That's where the Skynet is coming from is artificial learning and that's from his work They are it goes so deep. It's insane.. Tyler: There was an a so one of the things it's so... Karl has these meetings... you.. where's he from? What does he um, Julien: London, UK.. Okay, you can so yeah, he has these Monday meetings where I think like anybody can go ask questions, right? And One of the things that happens, I guess what's interesting about this. He has everyone has all their questions up front Yeah, and then he just dives in to all of them Julien: Yeah, and whatever I'm sure he wants to talk about.. And half the crowd leaves halfway through cuz they also so pissed because they don't get it... Tyler: Because they have no idea... Tyler: Yeah, the other side is I think he says these meetings are like 12:30 in the afternoon And then he will not say a word all day until he's there like the first word he speaks out of his mouth is actually at the at the thing which is crazy.. Julien: Yeah, so he has an understanding with his wife and kids He is a very interesting guy. Let's put it this way and where they are not to talk to him that morning at breakfast they grunt and go mhmhmhmhhh I mean, maybe they talk to him, but he does not answer back outside of grunts or stuff like that I don't think anybody talks to him until he shows up and he small it's not a meeting. It's It's more than that. He's in a what do you call that a theater? Tyler: Like a lecture hall Julien: Yeah, because he rushes he teaches at the number one college in London when it comes to neuroscience... He is the leading neuroscientist in the world. Tyler: Well, this was one of the notes I had was 90% of papers that have been published on brain imaging use Fristons work. Well, he's cited as yeah, but Julien: He is cited in most of neuro. He is the leading neuroscientist It's no question. He is cited on everything he is the Einstein of neuroscience Oh, actually I would I would I'm tempted to say Einstein was the Karl Friston of Physics He's produced Julien: Maybe not but basically they are the same level. They come up

with nouvel shit. Like no one else does They said he's produced a thousand papers since the year 2000 Julien: By the way lot of it is mathematics.. Tyler: And 85 in the year 2017 which is like four.. one every four days Julien: And you have to see the papers.. It's like, if people want to see there is one for example in 2007 called "free energy" in the brain Julien: This... The concept is the most innovative stuff that I've ever read since Schrödinger What is life? Tyler: Yeah Julein: Which is a direct continuation of Schrödinger's work. No. No, it's the most, the most brilliant men There is no such thing as brilliant men ever, but alive right now... ...there's no question. Like so in that theatre where he's giving his lecture cause he's part of a, you know he is a fellow Neuroscience University London, whatever it is. Like there's a name for it I just cant, I never remember But if you look in that theater you have.. a third of them are all artificial intelligence people.. The top in the world. So they understand what he's saying then you have neuroscientists, psychiatrists, you've people from.. The top minds in the world go there and again half of them leave the session saying: Well that makes no sense Tyler: Yeah, that's well, the first few times you had mentioned like the subject I could understand little pieces, you know, there was a there's a concept that we'll get into a lot of which is the prediction error thing. Fundamentally, I felt like I could figure that one out, because I could, you can almost directly apply most of it, a lot of the other stuff is.. it's almost though that a lot of that math is for validation or to, you know, not validation, but too Julien: It's logical explained Tyler: Right. It's to explain and I get into that and I'm like whatever this part is. I just don't need a ton of it..I need to Julien: But you have to understand, no one can understand Karl Friston. Tyler: Yeah.. Julien: Like it's it's one of the.. The stuff Julien: Like that's why half of them leave the stuff so pissed, because they're like, I don't understand what you're saying, but I think they're liking the .. where he starts. The finality versus causality? That's where free energy and the brain.. what it is about It's a direct continuation of Schrödingers' work. So we're gonna get into that.. ..but that's.. If you miss that part, it would be very hard to understand where he's coming from, I think. Tyler: Yeah, I agree So, where do we want to dive in at the very first into? What, what the work is? Yeah Julien: Well, okay, so he's a neuroscientist, right? But I think we have to start with a very quick with a Schrödinger "What is life?" Like I said.. People don't understand Schrödinger, he was the guy with the cat I mean,.. But he came up with a wave function, which is the most successful theory ever produced in a sense of: it was never proven wrong. Right. He is the one. The wave function is the probability of an electron to be in a certain place. Whatever everything is based on probability.. Welcome to quantum mechanics. This changed the world as we know it. There is no computer, there is no Iphone,

No - there's no TV, LCD or any of that without the wave function. Schrödinger's wave function. So, he is one of, he is one of the Einstein of Civilization which is.. well, most people just don't know who he is, outside of the cat thing. Yeah, but he is one of the greatest scientists ever. And he came up with a book in 1944 called "What is life?" Where he is trying to connect Biology and quantum mechanics. Saying like there is a deeper relationship that we understand with quantum mechanics, life and stuff like that And so.. That book generated the field of what is called quantum biology which is where, we were we're going and In that book he talks about causality versus finality. So, causality basically, the idea was that you have two types of things. You have non life and non-life Non life obeys the laws of causality, which is ..you know, entropy always increases. We're going from order to disorder...everything.. Tyler: Well essentially just constant decomposition.. Julien: Basically and.. we, you start in an orderly state and you move toward disorderly. Everything decomposes and at the end we are all dead. But, if people like that because it works to All of physics today is based on that. Because it's not alive and it's a linear progression. From order to disorder and that's That's Newton's laws of movement. for example Everything is based as what the math go tack tack tack tack tack and people love that because it's a linear world without surprises like You know where the earth is going to be six months from now And where were six months ago where it's going to be six months from now there is no time element in that linear system is to press to be called for call for There's no complication. The world is simple. We invented everything everything is safe. And that's basically the late 18-hundreds even or even Actually up to 1920 This was the world we lived in people don't understand but in the light 1901 actually someone said that everything that was worth inventing was invented because at the time like you have to understand like when Isaac Newton came up with the laws of movement we've thought Physics was figured out Few disabled few decimals out there, you know But up to the way that like a sixty symbols or whatever we had to figure out everything Here comes first Einstein right who come up with $E=mc^2$ know myself and then we go into quantum mechanics and quantum mechanics and you should introduces the idea of probability Like nothing is said and that's when you kind of when we kind of come to the realization at that point We had just we just left this phase in which we have discovered all that there is to discover somewhat right and to now it's like oh we don't actually know know anything anything because some of the stuff quantum mechanic was saying to This day are not possible and yet they happen all the time. Yeah, it's stuff like entanglement. Well two entangled particles No matter what. The distance is in between the two No matter what the distance

between the two are connected that goes against everything we know like literally if you take one particle, so particle spin in a certain direction right just to Crazy idea. It's called entanglement a particle spins either one way or the other if particular the entangled that means that want one particle if you You don't know which way they spin until you look at it. That's quantum mechanics converting to that one But until you you look at the particle, you don't know which way it spins once you look at a particle Let's say the one on the right spins Yeah, the other one will speed the other way. So clockwise counterclockwise like it's like basically I'm telling you I have socks of different color You don't know which foot orange is on which foot but the second my right foot is red as I see one I know it then you know what the other one is? Okay. It's the same idea right? So then you take those socks and You put them at the opposite side of the galaxy Right, so I don't know which one is which but the second I see I'll test One of those particles because that's the only way to know which way it spins Then I know that the other one spins the other way even though they're not Physically attached or in exactly physical relation. That's quantum entanglement I think yes exactly and we don't know what that means because it's a communication between the two that we cannot explain now the interesting I had actually just read a deal the other day where they had could somebody had confirmed Via what they had. Is there two Separate systems of observations. Yes, and while they would they would they would observe at the exact same thing at the exact same time Each observer observed it in a different exactly So not only can you have two separate? things that are entangled you can actually have the same thing and there was a problem because both Testing was right posted things were so but didn't they basically came up with different answers? So both sensors were right, even though they said the opposite thing on the same experiment. So With the point is we don't know what he means. Yeah, that's welcome to Quantum mechanics, we don't know but I think I don't know I still think that's an important thing to carry over into almost everything else that we run into. Is that there You can observe a very separate things and they can both be right for that moment that Problem is the state Is the state but okay so but the idea was so there was a probability That's what Einstein didn't like about quantum. Mechanics is the probability aspect and that's why you famously said God does not play dice Because everything was had a probability to it. Anyway, so quantum mechanics comes and basically changes physics so basically now Newton's laws of movement are Mostly, right? that mostly like, you know, that's entropy always increases or almost Exactly that change The world that change physics completely because then suddenly something that we knew we had that very line of view of the universe Everything is chemical.

Everything is that and here come quantum mechanics who basically change everything so Schrodinger after coming up with a wavefunction and the probabilities applies this to biology and it basically Separates things in two groups. No life obeys. The laws of basically newton's laws of movement as we go that way and Still on the quantum mechanical level is probability based, but then suddenly it says life or various different world. So Everything goes from order to disorder That's causality, but it seems life obeys finality which is trying to fight the disorder life does everything he can to not let entropy go a wire to not let entropy increase to because Maximum entropy is a death of a system anything that goes to maximum entropy has no interaction with everything is dead It doesn't move. There's no interaction. It's done which is where the universe if we are right will end up one day, right? so but basically life fights that does everything he can for entropy not to develop too much entropy is what It's called also free energy. And that's where the work of wisdom comes in handy Basically said it could quantify free and and that's his paper 207. For example, when it says free energy in the brain. We explained that Life not just earth just life in general he's trying to limit the Creation of entropy free energy. How does he do that by minimizing surprises that a surprise? Creates entropy creates disorder creates free energy. It's it's something that it's almost needs to be You know what? I mean, like now it has to be quantified and it has to be put together Otherwise we can experience more care. Yeah, exactly and so stuff was that's what so from a neuroscientist perspective That's what the human nervous system does it's a learning machine that has one job is to minimize surprises because if you minimize surprises you minimize the creation of Disorder you minimize entropy free energy and You can stay alive more so my question is then is Is an extreme on either side good in that department as far as what we look at what we're seeking for you know, you know, so we're gonna get into how we're gonna make this these predictions and the reduction of free energy for humans for the way We think the way we act the way we learn but is that it is reducing free energy Even though that's something that life seeks if we were to completely no you can do it But that's that was even further. What what he did. Basically is that so quantum mechanical moment for The study of life the problem is okay so there was Isaac Newton which was very like a real movement away and then quantum mechanics comes comes and Just changes physics right medicine did not have that the medical world is on the same thing. Everything is chemical Yes, just like physics had newton's laws which everything is line area. You do this you you do a you get B Medical science was like that with that empirical double-blind study was like look you do a you cause B You have causality everything follows a lining up Parian and even though it's not sure. Oh

That's basically what they wanted to get to because it allows you to have a world that is simple to understand that's humans for your Exactly an estimate because we were all the universe has no Responsibility toward you to make sense now This isn't this isn't seeking like a reduction in variables as though we're like crawling back into a cave. That's no no. No No, no, so what he did basically what Kristin did is he introduced the idea of probability? Okay, just like quantum mechanics said well, everything is a probability We know where the electron is going to be is going to be maybe But if I have enough of them I can tell you what the probability of it being here versus there, right? That's that that what Einstein hated that? Not enough numbers we know Can be anywhere we don't know until we look all right So that was the same thing with first and Kristin came about and said that so the belief was that your neuron are basically at rest until they being Excited being awakened by a stimulus from the outside that creates a reaction into your brain specially chemically so electrical so neurons creates a chemical reaction and Basically your body carries on so that we are a passive Reaction to whatever happens in the world the neurons don't do shit They're being stimulated by light or whatever then you react to it. And then you Carry on with the reaction to what happened in the world, which is my total understanding of the way that things work, right? This is the Newton equivalent Legion Rosamond equivalent for medical world. That's how the brain works right a causes B That's whoo. That's a it's a line in your world that can go forward or backwards that is simple to understand and where a causes B just like Newton laws like, you know the Apple falls because gravity don't Know right. So that's where we are in a medical world, right? Yeah, quantum mechanics comes about and say none of this is true with Newton's laws. It's far more complex of like shit It's mostly true Newton's laws like in most of what we do is true But the second you start to go at the very smallest not true anymore. All right So first one comes about and say the same thing like it's all about probabilities. So we introduced Quantum-mechanical moment. So what does he say? He said that that's not true. Your neurons are not waiting for anything They are constantly trying to predict the probability of an event so what does he mean by that an event is not based on the outcome is your system is not trying to predict if the Apple is going to fall you're trying is Your system is constantly trying to predict how it's going to feel when the want the Apple falls How you going to feel about that whenever you see the Apple falling and so he's gonna try to make a prediction Based on that the probability of this versus that like, oh there's a 60% chance I won't like this when it happens and then after that it's going to match that prediction versus the actual sensation of the stuff and If everything goes according to plan is great If it doesn't you have a prediction error, and then there are

corrections that need to be made Yeah, and that's what we're going to talk about. But basically he introduced the idea of probability like your system is never passive It doesn't wait for shit to happen and reacts on the contrary. It's very proactive and Calculates the probability of something happening constantly Something happening not being outcome based being how it's gonna feel and that's important thing I think is that and that's that's where I had to Get past the fact that the I thought in the beginning that the prediction was about Things like survive on things It really is only at only trying to match a feeling because I suppose it's how you become in tune with reality You become a tune with reality through your peripheral nervous system Which is the only way we have to connect with the outside so perception is reality, too It's like that's basically what is showing with this is to life Perception is reality To something not alive because it doesn't have perception. They have their reality. We have our early supper Yes, because we can feel and therefore we have our own quantum mechanical system based on probability Of how is something only feel like and you are constantly Anticipating the shit to happens next as a learning mechanism. So we are on alpha zero Like that's what artificial intelligence is going to world is it's not mimicking human intelligence and sense of consciousness It's mimicking human behavior in the sense of learning. Mm-hmm It's artificial intelligence is about learning like our nervous system rooms Which is probability based of how something is gonna feel? There was a so one of the examples that I've come across was like how did they word it? They said like the the The fear of pain yeah is actually far greater than the feeling because that's your mobility Because especially once you've experienced right if I put my hand on something that just burn the shit out of it. That sucks Yeah, but then when you go if I were to go back again a week later you get I get closed desk I will almost need somebody to stuff my hand though, because it's it's an intense experience though at this moment There's no fucking input. No, but the probability of pain is so high. Yeah your system go easy As it is what you know from that Yes, so your waist learn so you always make a probability of what happens and then you learn from what actually happens Yeah, so there's two phases There's a prediction then there's the expression you predict something and then you've got a new POTUS you're gonna make any purchases So yeah, and basically you're gonna confirm that it purchases with what the sensor input you get from the world outside if they don't match There's a problem is a collection if they match then you know for sure and then you move on with that feeling So that means that you're gonna have to redo it constantly before we go that when it when it matches That is what we would define as a reduction of free energy Right. Yes It's about matching not good and bad does it's not good or bad it's not it's not

this is the things beyond your control I like this who I don't have nothing to do. It didn't match what I thought it was going to match. So I I put my head on the stove and I burn myself like you can see that with a kid the next time they come Even if the stove is not on you can't get the hand close to it because you know, it's gonna hurt you're wrong but it doesn't matter because at that moment that's one of Mario's I would have to force like this and you go like half So there's actually nothing. That's why I can't get my dog in the fucking bath. Yes And then it's not for everyone because he's incapable of recreating To change his prediction. So that's we're going to talk about that So when is a prediction error, then they are different of dealing with it? But you have to understand and once you're sure of a prediction It's like that's how your system as Leon is moving forward with that. It's so hard to break Mm-hmm, because it matched so if something hurts, you know that for a fact he always hurts. Yeah, you know like the taste of something we're gonna have to associate and in it takes a lot to break that I remember there was a Suzanne old I think this is some old like military interrogation thing but there was they had actually said that electrocution was the thing that they had used because it's such a I Mean like massively intense sort of pain I never actually Didn't play - a few times - yeah, I've had a few moments working as a technician where I got Real bad and one of them - where like I almost didn't crawl out of the fucking situation But but so I know what that feels like And it feels really really really really bad. Obviously But but they said when they when they would test it when they would like to torture people one of the reasons they would use Electrocutions because it sucks so much, but they actually barely have to use it on you Meaning they hit you a couple times and then all they have to do is threaten because you're like no no no, no No, you can use it like like ah It's it's that's it's massive the urge to been away from this is important because that's the nature of chronic pain Yes. Yeah. Well, you don't need the stuff anymore. You're already hurt. Mm-hmm You see that with people I can't get away from pain anymore because that's the problem. Is that prediction get? Is true now and constantly active. Maybe this would be less This might be a good opportunity to to use that as an example like a specific example for news. Let's let's go down the Chronic pain route. Yep, because I think a lot of people will understand that. So let's say I deadlift And I have a thing. Let's say it's a physical thing. It's this at this point it's not manual right and and I I don't stay tight things shift a little bit and And and it disc gets a little wonky and also my back hurts a lot Yeah I'm gonna start walking around on it and I'm gonna start to relate certain movement patterns or positions that are gonna cause pain While it's reasonable at that, I have an injury, so I bend over fuck that hurts You know and eight days because everything I do I try to play

exactly in the plane actually the problem with this is you bend over and Ear hurts. That's not the problem the problem. Is you go. Oh I felt something and I'm fucked for three days Yeah, the next time you feel something anything Now you freak out because you're like last time I got fucked for three days. There's no greater fear than that one going like Oh I fuck my back. And this is when and in training you'll often see somebody who come in and they'll say okay I'm ready to start going but I'm not a deadlift and and that's when you do things that it's not to say this is The reason they're like I can't do it I can't like that's when you see the fucking team because they have learned that once they go clock It's three days. It's a three days of a fucking nightmare. And that's that we're going back to electrocution That's the same printing and so what happens is they will still end up making Because we say that the fear of pain is much greater pain that that that learning We have already done the learning the the prediction has matched at that moment So now they know so what I see that's always interesting like is some of these other parallels that still fit right? So one of the ways like, dr Stu McGill does some things with back injuries is he's like, well do what you can it does not hurt. Yeah, whatever That is do that if that's walking if that's walking and wiggling your arms if it's carrying weights start there But very often you'll still see people once they get back a little bit of intercession Yeah, first few and the problem is over the course of a rehab they don't they don't expect that There's going to be setbacks and there always are right you're gonna get stronger Yeah poorly once or even even if they reuse the same but the setbacks because they do barely anything. Mm-hmm, right? They go like oh you're fine but the problem is the second you go back into the position the problem that Caused the issue in the first place and she's you relax when you none of that is fixed and so all you got to do is go into a Deadlift position and not feel comfortable and then I'm gonna and you're not they lifting. Yeah, exactly. Yeah and And the bar is not coming off the floor. Exactly. Okay, exactly But but that's how you end up with and that's why from an imaging standpoint they always say that you can't tell someone who's in and someone is not because a lot of it is going to be how they've What they've done? Over the course of the actions that they've taken since then to allow them to be To not have this really negative prediction associated with this by the way That's part of the issue because that's when doctors will do an MRI And if you are bulging discs say you have a bulging disk, even though are this now Even though the bulging disk might not be where the pain or cause the pain or where the pain is because there's been many studies on this where Eighty ninety percent eighty something early eight percent saying it was of people out there with no back pain have balls in this L everybody does you're about in this but since we cannot Understand

where the pain comes from or we used to then it was I need a line Arial world where there is a cause a Visible cause for everything. Yeah, so I'm at B. There's a I take a radio I see bulging discs There's my a it's bad science, but it it happens all the time It's bad sense because we did not have a quantum mechanic moment. Where a Garlic fry Sun comes about and said, but that's not what's happening yeah, so that that that's how we can turn something like chronic pain into a self-fulfilling prophecy because that's the corner of life is a Self-fulfilling prophecy. The point of life is to limit is to mitigate surprises So literally this is my prediction and you're gonna do everything you can to make the prediction, right? Because if you if it's not right you have a surprise surprise creates free energy entropy that leads you to death eventually so the body doesn't care about Right and wrong. It doesn't care about good and bad. It cares about no surprises and this isn't even an and this is almost It seems to me separate from a survival or evolution thing. It really is a life versus not alive. I got areas Yeah, you know what? I mean it look like this is literally life behaves in different ways and things that are and that's what that's the depth of what Schrodinger was saying this is the most important concept I've ever read about What Shawn India said that's why I was like how he has not made a ripple or maybe a report in the world obviously because now this twisting but this was the most Mind-boggling thing I've ever read in my life past niche who to me was everything Where's that book from Schrodinger where I was like you guys don't understand what he's saying? He's saying that life as his own way of doing And that's what life. Does he fights entropy? so but that means on on a quantum mechanical level any single life will do everything he can to mitigate Surprises, that's how we do. That's how we move forward. That's what life is is a way to mitigate surprise So the proto the purpose of life is to fight death So we addressed in the chronic pain example where you have you have a negative reaction to something right? It's a prediction It's whatever but now but now you know, what your buddy Can you sit now you well, they're all so there isn't any for you, you know What's gonna happen if you get into that position the next day you're just no So but how do you how does one what happens when the prediction is wrong? Let's say I Promise if I give an example off the top of my head We're gonna be married to it for the rest of the thing. But well what's example you can think of? Okay easiest way Let's take away from training like for example Sexual behavior. Okay, you're sure you don't like something because you saw a movie a porn movie or what everyone you thought? Oh, I will not like that because you judged it from a belief respective not having done the stuff But you've been told parents friends or whatever. This is disgusting. I don't do that. Right so, you know going into it that you don't like it you happen to do the stuff or whatever and suddenly

the outcome does a find not the outcome, but What you feel? There's not much what you knew for sure you were going to feel that's easier to Get in touch with because that happens to everything. It's like that's not how I thought I would feel and now you're like wait what that's a surprise and now basically your own stem is like has to Again has to go against the word and has to marry the two back together So now once you have a prediction error, yes, this is how I'm going to feel well, that's not how I felt It was most whether you like it or not is basically the other way I'm gonna love this. Huh like you think like You know you oh I'm in love with that guy sex is gonna be awesome Not always right or the opposite anyway now you have you have a prediction arrow these three ways to correct that you can it's There's a balance between the three and usually dysfunction happens when it's one over the other anyway, we're gonna get into that So the first the first thing is you have to change your prediction Right, that's the easiest way to do so, I'm on the sled I'm gonna do the sled I'm gonna do 150 in 12 seconds. All right, I put too much weight on it It's not gonna move. Yeah, you get or you get two seconds in you're like No, this is not gonna take 12 seconds because you fucked up on the way. Yeah, right So it's the same idea as you thought the sexual position was going to do that and you're wrong Wrong prediction. All right, so you have to change your prediction on that You only do 1 2 3 So let's go back And so let's just say that on the one from a so basically where we change the prediction, right? From the from the training standpoint it is kind of simple and they're like, okay. Well now my expectation is very different Yes But exactly but this is a very important point about that how you make but you have to understand where you make the prediction this Is where your brain comes in the brain? The purpose of the brain is to allow you to make prediction, right? so there's certain parts of the body that are designed for these three ways to marry, the Prediction and then the expression, right? And making the prediction that's the role of the brain like literally I have a study if you all want that I can link about all this but actually it seems that That's what the interoception all that stuff comes in is the role of the brain is to make the prediction This is what the testing I achieve that I talk about all the time is about is that's the Job of the brain. We thought the job of the brain was basically again The neurons are down there just passive and then they react to stimulation and the order of the body to do stuff. Nope That's not what the brain is for The brain is to make a prediction is to decide the environment you're going to be in so that's what the test negative is So that means that's also where you sense of self is. So that's basically you're gonna make prediction based on who you are Who you think you are on your personality to change your prediction? You need to change who you are So what do I mean by that on the sled? I put too much weight Cuz

I thought I was the guy Capable of pushing that you're not that guy but you're not that guy so you're gonna have to change your sense of self because your sense of self is the one Deciding the prediction why because it's in a test negative network and a testing at your network is sense of self Morals MX prediction on past and future it's where the any change in in the peripheral stimuli So this is where the test negative Network is so important that we talked about in other podcast is that's the one responsible for making the prediction. That's who you are Who you think you are? Right, so to change a prediction you need to change who you are This is why addicts in order to get away. I need to change who they are John does drugs could John is with his friends John as an environment They're right. In order to take go away from drugs if he's addicted enough. John has to die and become Mike Yeah, cuz Mike want with drugs, but every time I go back to being John in that environment I'll go back to drugs its environment behavior identity Right identity is based on is is prediction prediction is identity. So on the certainly inverse of that if If I go up and I'm gonna I'm gonna try and deadlift 650 pounds and I think all right I think I can get this it's gonna be and then it comes off the floor like it's nothing. Yeah Next thing. I know I'm like wow Seven and a quarters fucking not now. I'm the guy who can have to do that Well, you know what? You know, why because you become a guy who destroys 650 Yeah, and so now you want to become the guy with a lift 7 Yeah, like that's you know, like so as if I were to go anything this was going you know I mean you go to lift that you think is going to be very challenging Yeah, and it's easy all of a sudden you're like, maybe I'm a bad motherfucker exactly your sense of self point up Yeah, there's something where your previous 90 percent today Feels like a fucking 120 percent. Yeah, it's like a Reality check today your sense of self goes down So because you are you are the prediction in that sense, but neurologically speaking you want a prediction That's the problem when you want to do 85% today in training no matter what is if today for other reasons happen to be heavy Yeah, it's going to crash your sense of self Because you are the prediction from a neurological perspective from a testing achieve network You are your prediction you have to lessen that there's different parts of your body There's different parts of your nervous system is different one This is the central nervous system this a peripheral there's basically the enteric nervous system There are different parts of you. The one that makes the prediction. That's the goal of the brain That's the testing a team Network and that's where your sense of self is. So that's why making the wrong prediction can be so Damaging to you because it directly links to your sense of self from a neurological perspective because your identity is Tied up in the prediction because that's how you see everything Because that's the way works. Yes. Well,

okay. I am not about to judge evolution 300,000 years to become human the nervous system has been around for five hundred million years. That's how it works The test- network is the reason we on top of the food chain, right we build things In society, you know, basically the one thing we have as humans over every other animal is we can change our environment better Oh, yeah, so all testing achieve a way of making prediction is better than the others we better on making predictions So a sense of self is superior to every other anymore because we can change our environment Better than others animals, its environment behavior identity tack tack tack That's what I call a crystal environment is basically one when you can when you decide if you're going to be successful or not But that's a very important part when we go back Go back to pain right because you have to be successful in in making your prediction. So There's a problem with addiction and pain and everything. If you know for a fact that your prediction is dead left Equals back pain then when they lift equals back pain it raises your sense of self. Just like for you Oh I can I smoke six fifty I feel better Yeah, every time I hurt myself on the left my sense of self goes up because I was right in my prediction Fucked up he can be though. Yeah, he can work both ways Yeah, and then they become even more certain but then he goes the other way That means if that lift doesn't hurt my back then Who am I? Because I'm wrong in the prediction. That's a surprise what you're trying to fight against So that's the power of the prediction and what happens when? There's a prediction error. It's to life. It means literally death where your choice so if you make the wrong prediction Whether the future is good or bad doesn't matter That's the problem is you make do as long as you make a prediction. That is right. Your buddy's like awesome So we're trying to constantly basically just address everything with certainty because certainty is order certainties life Yeah, and then and then as soon as there's something that's a surprise like that's fucking chaos And that's why that's why it's there. There has to be a correction. There has to be a correction. So that's the keys There are three Corrections if you stock in one like changing your prediction your sense of self is attached to that That means that every mistake that you make any super non mistake any Surprise that you have in your life for better or worse Diminishes your sense. Well, it's always bad. So any surprise even if it's a good surprise? It's bad if it's a bad surprise It's bad in the sense of any surprise is bad. If you can only this way ways to correct The first one is changing the prediction if that's your way to correct errors, then any surprise is bad Even in stuff surprise is has a good outcome Outside you will still see it as bad and it gets worse no matter what how good that surprise Is it lessens you as a human being it makes you feel worse about yourself? That's why in some people that are fucked up happiness hurts

because they're happy that's a surprise that surprise makes them feel lesser as human being because Really it does Just destroy the sense of self because it is a surprise because that is the only way they can deal with a prediction error that's the problem and that's when you see a lot of issues with ego and stuff like that because if you don't have the other ways this prediction error then everything is a challenge to your sense of self any surprised so Anything that feels good when you're sure it doesn't lessens your sense of self. Imagine the hell that you live in Imagine how scared you would be to walk around knowing that everything that feels good Will make you feel like shit as a human being that is help that is literally hell on earth Were you raised Catholic or no? I'm A grandfather. I'll show a lot. My grandfather thinks he raised me Catholic I can tell you that But but so we have the the changing the prediction is the big one and that's a big problem, right? So yourself, what are what are your other arm? So the second one is you're going to create movement to basically make the world fit the prediction Right. So that's the one where they live hurts my back, right so I'm gonna fuck up the phone Like I deadlift and it feels good. I don't want that so I can either change the prediction getting like one Maybe they live doesn't fuck me up or because I don't like it I'm gonna change my phone until I hurt myself. That's self sabotage. Yeah, that's you all the time But again self sabotage in itself isn't bad. It's just trying to limit the surprise so you could basically change the movement Into one where you hurt yourself and then the prediction came true and that's okay. So that's where also like I'm gonna freak out. So basically you will do movements to are freaking out which can be Mouse breathing Which basically you're gonna you're gonna act on the prediction to make the world fit it? like so The position doesn't come true. I can either change my prediction or I can make the world fit the prediction I can create movements that will make me feel the way I thought I should So you can go the other way, which means I love to squirt. I'm gonna change my position until I feel the squat Feel good. Yeah like for so what is my body? you you love to snatch because at one time you snatch and you went oh Yeah, that was awesome so the way now if you're snatching feels like So you have a choice you can quit snatching or you can do it until you feel that movement and you go That one you find that again that one Yeah, and you know what you did by the way, but you know, it's that one that that one was true That's the one right and that's so that's what we talk about technique about anecdotal evidence about learning skill because that's what learning skill is It's number two. It's Learning to make something feel the way it should so that's where mastery of movement comes in that's where my story that's where my Mastery of making espresso come in is the action should feel a certain way Then you start to go through repetition again and again and

again and again because every single repetition gets you a little bit closer To what you think that action should feel like and that's where mastery comes in That's why after a while because then eventually you have very precise control You have moved Exactly, and that's what people understand how masters of anything can be so precise in what they do. How can they rub duplicate and replicate? something to almost perfection How can the guy who makes the soup makes it and we are so slightly different, but every time it's great Right because he has mastered the movements Necessary to duplicate exactly that so that's what you see the great Olympic weightlifter Where they do a movement that to the untrained eye always looks the same. Yeah It's is that and you go like oh my God, look at you. Can you can tell the mastery of movement? Why because they can duplicate and replicate the movement almost To the same level every Tama works for them because that number two every time there's a prediction error We didn't feel right they'll they have learned to adjust To make the bow do what they want well and as they progress, you know Same thing with like the sport of basketball if you yeah, you see the three-point shooter. Yep I'm not one of those but but like but but when I was shoot still like like, you know, I would say within Eighty percent eighty percent of the time the ball leaves my hand. I Would know for sure if it's going in right and mastery is Kobe Bryant knew 95% of the job. If not, ninety eight Yeah, and then the way that he could get to that feeling exactly and the greatest shooters out ninety-five the time nailed it That's what the guy turns around a Stephen Curry. He knows. Yeah. So him. Is that only 99 percent? Well, he knows that when he feels a certain way, he makes it that's mastery of movement that's repeated That's because every single prediction area it physically physically worked out Change the world to fit what he wanted That's interesting and that's the power of that's mastery right there that's anecdotal evidence That's where I will talk about this but that's where I fight for anecdotal evidence so much is because that's where mastery is That's where you cannot learn a skill with anecdotal evidence because they have a whole thing about any little evidence being Being useless and that's completely wrong in the sense of you cannot learn any skill with anecdotal evidence So you can't use any totally villains to convince others But that's not the point. The point is you need to convince yourself first. Yeah I mean I could read all of these studies specifically on how a basketball shot is supposed to look Until I until you have to get in there ago and I and I know what that feels like, it's like well, yes And then it applies to me. Mm-hm But that's what matters to me That's the only way I'm gonna learn the shot Exactly and but from an object from a Freeston perspective that is extremely important because it's one of the three So you have to make the correct prediction? Which means so you have to that's what

memorization comes in in that sense of this is what a shot is But at some point it takes the fucking million shots integra, what is anecdotal evidence of you? Basically going through prediction errors physically and many times you like or gonna quit because you don't want to go to that basically second way of Fixing prediction error. Yes quitting is I made a wrong prediction, which is Shooting which means I want to play basketball So you're gonna change your position going like I don't wanna play basketball anymore because you're tired of going through the second solution Which is changing the world every single time until if it's what you want. Yeah so Third option. Yeah, if I have to change the predict the prediction is wrong. Yeah what before we get into the third one? This is what I want to know. Are these choice based? Necessarily is this a decision to choose which of these three it's it's a balance. But yeah, do you well Now the question is who makes a choice, right? Is it it's basically from you have learned from previous behavior Which one is more effective? Okay, that's why if you have you go to if you quit long enough It's obviously effective because you change the prediction every time then it becomes your behavior so environment to behavior to identity if You have decided to change your prediction. You create an environment where anything goes hard you quit because that's the environment you create it because that's the way to that's how you change the That's how you fix the prediction error by changing the prediction every time so that will do be your behavior Which is every time something that's hard. It's every time this is surprise. Yeah, I change the prediction that becomes your personality So you create an environment where things get hard. It's a surprise. Yeah you much rather quit then you become a quitter. So it's environment Behavior identity what you get to choose is the environment? Okay at that stage you get to decide I am NOT. I will not If this is surprise, I will deal with it and not get away from it that will affect your behavior You're not a quitter anymore, but you can't control the behavior. You can't control your identity. That's what people quit but see another query Oh, no, you don't get to do that But the only thing you can desire you can decide of you can choose is your environment so you get to choose which one of the three of all three you get to use that's where this comes in or Consciousness, whatever the fuck it is, whatever we have that all weather animals. Don't have we get to create an environment? Lions are stuck They're not going to go migrate 300 miles to find this basically they have a territory and they are stuck down. There's no food They might move a little bit but basically therefore they're not going to build an air conditioning coach this to her they're not going to be able they go well in so Well we can do us is we are very good at ads with the prediction thing We can create an environment better than anybody else. So that's where I think People need to understand where

the success in life is is they get to choose the environment. They don't get to choose our behavior They don't get to choose because that is based on the environment environment behavior identity But you get to choose the behavior That is the one thing you get to do is you get to go like this is where I want to be so You think you're fat you're out of shape. You're unhappy you decide to go to the gym Now once you're in the gym, you don't get to decide the rest. That's the problem, too Is you go to the jinnah? You think it's gonna feel good you won't know and so now basically from that you say Oh, I'm a loser. You're not just it's supposed to feel like shit You're just wrong in your prediction on how workout is supposed to be. It's not supposed to feel good It's supposed to make you better or better than supposed to so after that It's out of your control But being in that gym been a correct gym will make you train harder, and then eventually you'll become a winner But you don't get to control that part. All you get to do is choose a gym. So what do I mean by that? You join a gym, but you don't like the mood Right the drive doesn't fit you. You don't like the people there. They're not supportive to you or whatever Do what do you think the best? Program in the world will do to you in that environment nothing Absolutely zero. So now you sure you went into that gym You chose that gym, right at least you chose to go to a gym. That's the gym you're in now What you would like is you would like to change the coaching There you would like for people to be more supportive and everything. Is it? Yeah, but it's not that gym Mm-hmm, you don't get to choose that and so from there after that basically It starts to text your identity because you're like people should be like this. Oh now you bitch about the facts So now basically you're making a prediction based on. Yeah, and you go like I can't do this Yeah, you're right. But because you trying to control what is not controllable? What is controllable is? Change the gym until you find one that fits what you want and once you find a gym that fits what you want your behavior Will be of someone who trains harder because you feed on everything and from there. You become a winner So the wrong regime will make you loser. You don't get to control that And then you start to associate different things with different results, too Once you become a loser, then that's a prediction you make so now it's a feedback loop, right? And the prediction is I lose and now you lose and now you're in a bad place where because you try to control Far2 down like the only thing you can control is the gym your gym makes you lose or find change the gym That's the only thing can control these that is your environment Yeah, it's interesting, you know Like so if a regular person walks into a gym and they see some big Thick next son of a bitch dead lifting and banging weights and that guy's an asshole To them from then on out that person going into the day. I'm seeing any big strong It's gonna be like

fuck this place that tells you two things cause people ask us all the time harsh ride to my gym Choose my gym like in jujitsu or whatever. I'm like get in there. See if you want to go back. Yeah, do you feed off? whatever Energy, is there. Yeah if you do You'll win if you don't you'll lose and there's nothing you can do about it Because you're not gonna change that I saw banging the weight he might change but at is not under your control And you telling him he's an asshole Most likely won't change in my way but you know and on the other side of if you go in and there's some big thick necks son of a bitch dead lifting and Bang a bunch of weights and and then he talks to you afterwards and he's a nice guy you have no problem same situation happens exactly and you're like you're like oh by the way That tells you an important you a gym owners are out there because that means literally you Defining the outcome of that person's life by providing an inviting environment or on one that is non inviting That's what to meet the gym owners we go back to conversation We had so many times are so important and I think that's the only job Really is create an environment where the kind of people you want in your jib can thrive? Yeah, that's it. If you create an environment, you change the behavior you change the behavior you change the identity That's how much power we have in people's lives So if you want your gym to do better all your way Change the environment make the environment into something that will create the right behavior. That will create the right identity That's what I think the gym is the most important place in the world. Yeah so our third option, ah This is actually the one yeah that I actually I walked over at seven or eight times and it was completely misunder That's so this one this one I'm gonna need you to kind of guide you saying I'm gonna ask a lot of questions from this as in because I Don't fully understand this one. This one is basically we have to cheat in order to change or correct a prediction error We have to change the way the brain samples input. Yes. So basically is that's a third way you could change The area is you could change how you feel about something. Okay? Alright, so that's the source of most dysfunction right The again the point is to mitigate surprises the surprise happens not good. The faggot surprise happen is always bad So I need to weather change the way when I think about make the prediction, right or Move the thing until it is what I want in the feeling sense, or I change the way I feel about it that could be a That's where you see in a life changing event like a near-death experience We are basically they used to have a certain personality and everything. They used to feel a certain way about These things that used to be important to them no longer no longer are because they have changed the way they perceive the world Right because that's what happens to die. No funeral in any way all I stopped. So that would be the source of dysfunction This will be like the source of

schizophrenia or stuff like that. Well something happens and basically, I'm gonna And the son of Jesus Christ right? I'm being was proven that no, you're not right doesn't matter You're not going to change your prediction now, you gotta you can't make the world The world's that could get on board with that all exactly oh and you can't walk on water you can't change The the world in the way that proves your production and so what you're gonna do The only thing you can do is change that basically how you feel about about things That's where you could see schizophrenia coming in basically, so you have change the expression so that would be the source of most dysfunction like you create so Psychosomatic events. Yes, that would that would be that for example, so the deadlift is gonna hurt my back I left no, it didn't my form. I come managed to My form hasn't change, right so I did not physically hurt myself Aggravation nothing yet. I wake up the next morning and I'm in pain Because I made the world the prediction fit for example, that's one example But this many but that's worse psychosomatic events will apply easy. You have made yourself feel a certain way about that So there's a positive as well. Let's go back to our sexual behavior Yeah, there is that action that I thought I would hate but then basically this is surprise that tells me that this was not bad It can be because I never wanted to do this cuz this is morally bad But the men I love all the women. I love relax that action in bed Alright, so I have two changes. I can change the position I can basically Not do it or whatever. Oh, I can suddenly the next time I do it. I can make myself like it. Yeah, I Can change the way I feel about that? That's what the psychosomatic stuff happens so much. We see it with placebo Stuff all the time. We see it all the time and that but that's also in fortunately the source of many dysfunctions Because that's what also chronic pain is is I am so sure that something hurts me that it actually does. Mmm-hmm So this is people getting really good voices going like oh my god My finger hurts so bad. They're not kidding. They have trained themselves into feeling pain of anything even though You could say oh I poked you, even if it's not true, and they would actually feel pain That's the source of many many dysfunctions What? I'm trying to come up with some other so for this for this third one because this is the one that needs Clarification the easiest way to do this is the thing that we think would fall into a lot With food you had mentioned. Yes. Alright, so easiest way to let's say I want to go sympathetic Yeah, I enter a situation where I know I need to move Right. I'm bored. Yeah, right. I'm sitting here. I'm at my desk and I'm feeling bored. But basically the need of it's a change of behavior right I need to do some And I need I need I'm angry. So I need to fucking express the anger, right? So not changing the The prediction because in that case your body means you feel you need to do something So ever you start

enjoying what you're doing you want to work and you don't want to do that. So second you need to go Sympathetic, but you don't do that either sympathetic can be mouse breathing or Basically you need number option number three, which is I need to feel Sympathetic. Yeah So instead of acting I could go grab nuts something that would any food, honey sugar that will create a sympathetic reaction because now I'm at work and instead of changing my prediction instead of Making me like changing my work into something I would like all I need to do now that's option Number three is to create a sympathetic feeling and the easiest way to do that is to have sugar because sugar Triggers an sympathetic reaction. So now imagine a fucked up. That would be if Anytime I need to have a sympathetic reaction I could just feel sympathetic That means that anytime I need to take action I couldn't stand rely on something that makes me feel sympathetic that would basically be the third way To deal with a prediction error. So imagine if I put all my energy into option number three every single app So it's a triangle but instead I'm going to use number three, which means any time there's a prediction error I'm gonna I'm gonna use something that makes me feel a certain way so imagine if you could do that with food any time, I need to go sympathetic anytime I need to Basically make the world fit what I want and everything instead I'm gonna grab sugar that makes me feel sympathetic and therefore allows me to fix the prediction error Imagine the power that would have and that's our relationship with food. So we are using food as a third option Basically not have to do shit. So that means that we are using food Not to change a prediction even when we're wrong and not to do anything about it that is half had a Relationship with food is unfortunately neurologically speaking It allows us to not do shit well, that's I hadn't know here even from that from the thing I had read earlier was they said that he said that basically Both your perception your perception is completely bound to Action. Yeah, there's no there. There is no way around it Like like if I want to if I want to change my perception of the world Really has to be actually I've breathed if I have to move around my eyes need to move so I can see something different I have to breathe in in order to smell something there There has to be an actual action in order for us to like in order for your perception and so eating Becomes the biggest the biggest the easiest way to fix a prediction error Why because five thousand years ago did not have access to all those different foods Yeah, you didn't you had basically one type of carbs, right? Three types of protein and that was it so because there was so little choices of food They had to be so much more behavior changes behavior changes being the change of prediction or the making the world fit So now we have now that we have the quantity of food, we desire and choices of food We never had before it's basically limiting completely the need

for change of behavior or a need for movement yeah, so All choices of food literally determines how much movement and prediction we're going to do in life when we have at our disposal such a Wide variety of foods, which is the worst in human being frame wide variety of things It's really fucking good and quantity. And so it's always new and novel if you want it. So you're born You're like I can go have myself a new experience well all the food that created that feeling that one time exactly and And so that's the thing is I think you're either in most cases. You have the ability to really You could just replace a lot of necessary things with still is enough variety So you're not stuck in this nobody look at look at but it is that means that I make the wrong prediction Which means are like assholes know like the girl who's always dealing that Everything so I thought it was good. He's a dick. Yeah liked it. Thanks guys before you all you dating that bitch and everything Yeah, well guess what the other ten they were exactly the same way, right? It works just as well for women or men by the way Instead of changing the prediction which me realizing that your sense of self That you're wrong into you who you wanna Your personality is in the wrong place you're dating people that are destructive to you You're percent is a wrong place instead of changing that Instead of actually trying to date a good guy. All you gotta do is eat It's a specific type of food and you will have mitigated the prediction error yeah, and and because then the grand scheme of things food is still very short-term that becomes a lot of Constant behavior. So you want a cheese and you we need the different kind and we have both and then we get into the Basically you can see the connection between food and mental illness you can see food as a mental illness and I do believe that's what it is now food as because of the quantities and the choices we have is what's stopping you from Moving forward in life and that's why depression and food. We see that through the enteric nervous system There is a link between the gut flora Through the afferent nervous system the enteric nervous system going into it There's a link between the gut flora and depression because those choices in food will allow you will have direct impact on your behavior whether it is making the world fit what you want or changing your prediction, so Imagine how bad our relationship in the Western world is with food. Yeah, so that means that in order to Do you know I know a few people? Basically that there are stuck in life because the con change their behavior and they can't do shit. They can't find the hammies They can't exercise the way they want they can't find intensity because they cannot move to make the world fit what they want or They can change the prediction which is their stock in life Because in to change the prediction error, they go to what food This is the biggest drug and the most dangerous one of all because this is the one that will ruin your life and it's

interesting and back to kind of the the first way to Correct. The prediction to is there a lot of those people aren't willing to change their Sense of self either because they you know what, I mean So then you end up with with it another good option and then they're still in the middle of this Self-destructive or at least not so that's what basically drugs out by the way, it's drugs makes you feel a certain way, right? So That's what that's the source of dysfunction. That's the option number three. This is something with drugs because I mean the bad My life is bad. Yeah, right. I don't know shit about it, but I feel good I'll mitigate the error and often in the case with drugs is even though you don't feel good. It stills familiar You know what? I mean? So it's like oh it submits amidst all this mess like it's the right prediction I know I know what I'm getting is no surprises when you're on drugs everything feels a certain way So it's white drugs are so powerful but people have to understand food is exactly the same The difference is you don't have just not cocaine Yeah, but you have to gotta eat and that was always the thing that I thought was interesting with, you know? like habitual overeating and things like that people with with eating disorders was there that was the thing they always mention it was it's like yeah, but you know You don't have to like yep, you know snort something, you know Whiskey day like like but I gotta eat some food at some point So it's like you're always having to be in the environment with which yeah, so you're gonna so but it also means that there's a direct correlation between How many choices of food you have versus how many behaviors you can take the more choices of food? You have the less? behavioral Changes you will have it's you will either choose new food or your new behavior So if you want to do stuff in life if you want to change prediction You're gonna have to lower the quantity and choices of food that you have. So the most bowing in that sense Your diet is the better your life will be that's literally literally so on the nutrition perspective, that means that you have breakfast Basically, that's well You see that a lot with TANF you're doing vertical It is you want everything to you want to be able to behave the way you want to only be able to train the way? You want you want to be who you want to be then that means that? Especially the strong faith means this breakfast is gonna be a certain kind of fad that there's good to you the river that is yeah lunch could be a certain type of calm that was good to you from its rise and Nine-time is going to be certain kind of protein that does you good Right and you do that and basically the same choices then they art We allow you to have better actions in your life better decision-making That's a prediction and better actions toward your life You are using food to avoid better decision-making and better actions That's literally how we use food. That is it is that photo? I Think that has us that's a great conclusion, but that's basically the Freeston that's what That's

the conclusion. That's the direct conclusion of kalfus turns work That's why I created the the strong feed nutrition protocol is to fit Kalfas turns and shown in girls work. That's causality reassess finality But people have to understand the deep deep Relationship between food and behavior and even from a neurological perspective Then we go into the gut flora Which is another one on top of that Food and behavior are linked farm on and people understand and it is I believe in Western world the most destructive force You do not understand what that cheat day on the weekend does to you know neurological they don't like this I'm not saying to go cold turkey like this, but you have to understand that what you eat Will change your behavior on decision-making in life and not from a chemical perspective? From a Christian perspective. That's the problem is most people who can it. We also record. We only look at food via chemistry anymore Exactly. Yeah, and so they go like well who's to say protein? Does that stack? That's not that's even how I see Like eating dysfunction addressed. Mm-hmm is its brought as well? You see so you eat this because then you get this feeling of happy and that's this Chemical that Meshuga of does insulin this and saline that and everything that's a new Turns way of looking at the world. I said lining or way of looking at the world Well, it doesn't work like that. The world is quantum mechanical first. The work is neurologically speaking You follow the first a lot of all those other things fall in line just to make sure that correcting prediction errors happens Yeah, that's option. Number two also, like making the world fit to make the world That's what the chemistry comes in. Right but it's a what it's not a why it's a how not a why right? What matters is the why and why is the fist and model and so that's why Many times your diet fucks you up or different diet creates certain results because fundamentally though It's the same principle every time it's afraid it's the Freeston environment. That's what matters now to go forward This is not the format for us to give out any like good follow-up reading information for for instance work But I will adilyn. I will I will add actually I have one there's there's one really good concise I won't say good for laymen But it's a good description of who he is as background and and a good overview of some of the work and I'll add that into our link on our Instagram stuff in The brain yes, seven PDF. Yeah, add that one More I can always yeah, so I've been together a couple of those on it link tree and stuff So if you guys want to take a little bit of a dive I do what you can so but just to finish all this this is why I believe what we do is so important because people Have no idea how arrogant I am like they think I'm going to like do they think I know for sure that the diet is going to help sleep and Get for me leaner or stuff like that And then so they're all getting pissed at me, but you have no idea how far can actually I think I'm so much more arrogant and

they say I Believe some movement and nutrition we can have the controller of two and three and so I believe we can go at mental illness I believe we can go at addiction I believe we can do through that we can control so much. Most of this field called psycho neurobiology Yeah that goes into in a way the free stone model even though they're not talking about it Which means for whatever reason maybe they don't want to quote him, but you know obviously seems to be the foundational basis for it Sorry, just Google cycle neurobiology. It's obviously part of the feasting model But anyway, they're going at it through worlds through seeketh psychotherapy like the further way. That's the testing achieved That's so changing the prediction. That's the option number one They are not going into two or three which is very strange to me and I think we was Strong Fitness what we can do. Well, that's and that's interesting you had mentioned. It was really interesting how How you had kind of gone through some of this stuff to a strong fit even without maybe having this framework? in front of you to define it as You know, the first thing was going to be to change the prediction which is going to be basically Intensity. Yeah. Oh, yeah. So in in a way I've been doing this forever this is why the more twelve into this the more I was amazed how all right I was I was amazed at the power of a list like for example, if you look at changing Changing the prediction that's so as I was saying is changing who you are your personality, right? That's boom the questions It's you think it's going to do that what we're going to go past that we're going to change your prediction and to do that I'm gonna burn your sense of self. I'm gonna give you a new personality. Yeah, right So burn your soul until you know Bill who you are all that stuff that that was exactly that it was Changing the prediction. Well, that's exactly what Bill no question. That's that's that's the thing. We've talked many times the person who could do this Become the person we can take that sweater and over there be and that is the choice you create that you get to choose Now environment you get you get to choose the environment So for me, where's the harness where I choose the environment in which I could change? Me. Yeah, and the second is then just like here It's implement its mastery of movement movement quality being deliberate. You know what I mean? All of that stuff and number three is the nutrition is to stop using that That we are developing mental illness with the food thing. I Truly believe that and all the stuff I've read about the gut flora shows me the same the behavior the Western world is basically I think that's where the anxiety the depression Comes from is we are creating the wrong behavior and food is a major major warning to that. So You know in a weird way, that's what strong fit has always been about is to basically give control of option one two And three yeah is to give you the capacity to do all of it to give you the capacity to Not create free energy

to capacity to be alive now for you having had kind of Had these as principles maybe not worded in this way or frame with hotties or wise I knew but I just want to know I want to know what is it like when then when you have you're having to read? Something from you know, a pretty brilliant guy and all of a sudden it lives up here like oh fuck. Yeah that's basically what I have anyone is that Like what you know because I really stuff and I'm like, hey, that's my life. Yeah, so I am NOT crazy Nobody's it's - it's the vindication of every single ever worked on It's it's not even a vindication because I never felt that I had to be vindicated because I knew I was right I always know when I'm right, but it was the There is the physical Proof of my life's Work gold dreams, whatever you want to call it. There's a guy who quantified it in a scientific way showing everything you always worked on is Quantifiable in in that sense. He has mathematical stuff and everything is like, yeah, the Only all the stuff should be working on has a neurological basis You just felt it. Yeah And to read that on paper in front of you. It's like the guy talks to you directly Yeah, like you is when you read those books and you feel like the author is talking to you reading at first and stuff I was like Thank you And you were never chasing the never the math either. No like that was never like you just want to understand it I do. You know they what I said when I read Schrodinger, I was like, oh my god This is mind-blowing because I knew this when instantly the guys telling me no, no quantum. Mechanically speaking. That's what happens I was like I knew it and then I would still have like oh my god But then also they giving me a road map on how to do it now because I felt the stuff and I knew a lot Of it had to do it on myself not even all of it But I was like those are thing I could feel except I was like, I don't know how to do it and then there's a road map and to everything that is doing there's how save mankind and I think its name and it's important because you still think differently and Operate differently than than most of us. Yes, you know what I mean? which means you were never get you never gave a fuck either way if there was a Thousand research papers published on this subject or it didn't matter to you at all. No, because I knew sooner or later Yeah, but but it is but it is good to see Like even for me like like going through the few new notes from we had today on first and stuff My understanding of other than just the real basics now. My understanding is much broader Simply because I found different words to attach it to you know to it Which which head has to be at least? Otherwise, you're not getting out And so you you're trying to communicate you're feeling all this time. Yeah on it. And and now That's that I think the important part of having Kind of discovered that this work has been going on. Is that not now now there's an easier way to teach this to people Yeah, because I thought she was almost I thought my only recourse for

the longest time was philosophy. Yeah Logical evidence. That's the only way I could ever explain what I was feeling like guys I know I'm right but let me take you to philosophy who is blamed Because because I can if I'm good at something as I can see pattern so I saw patterns in because I read everything You know from paleontology to philosophy to history to stuff like that. So I saw the problems I was like, but I did it Plato and Socrates To help me explain it to people. Whereas no. No this dude. Is that smart? He actually figured out from in your as a neuroscientist if you know how this shit works This is His main kind and I don't hate me but school Jesus Christ He's a second. This was this was the really interesting thing that I I had circled this here on on my thing. He had said basically if it was to define his work, right and it was like He's trying to organize the principles of all life and intelligence And and again the quote that it that is left here is if you are alive at all you have life behaviors then must use show and that's where all of this where all of this is coming coming in - Exactly. So like life behaves in this way. Okay, so the guy has mapped out life That's literally what is done and is telling us there is a way to live because everybody is an opinion No life does no. No, it seems that life has guidelines just like physics Has the wave equation actually has the wave equation to define the behavior of particles Freeston define the behavior of life and is telling us in order to go to one more life This is how you behave. So it's not a matter of opinion anymore in the sense of that I lived my life the way I want No, you live your death the way you want and in most cases and to tie it back to learning what's really interesting is the the fact that I don't think of how to put this this is one knew I was gonna have a couple of these moments on this podcast right these Count no one understands it right. It's all good But but when we get into learning, yep in general, right, the first phase is memorization anyways and so a lot of these people and me included is the the concept of prediction prediction errors these sell for the prophecy model you know and and now on how Minimizing prediction errors and correcting them it's done Now that you know that now you can go about your day and you can kind of see where things are You you you you're in that memorization phase now and you can understand that It's not all here that at some point he's gonna have to be your movement Yeah, and then once you understand it then you can now Live in it. That's what as a musician, you know at some point. You're gonna have to feel the music exactly So it's not enough to know the notes at some point. You're gonna have to hear the notes. Yeah, that's Changing how you feel about something. It's sensory input. It's not enough to go tinkling ding-ding-ding. No, it has to feel a certain way So now you're gonna do ting ting ting ting ting So you have one two and three right now, you know that the prediction isn't good enough because it still surprises

So, okay So then there's a mechanic but then after you're gonna move Make an equally differently to make a better note and then you're gonna move based on how the note fields And then are you liked anything? That's An expression in playing music and that's the reason you play music Is that and now you're mastering it because you have all sweet going every time you need to change a prediction slightly because it's not What you thought then you make your fingers move a certain way and then you make the note feel a certain way And now you have music. Yeah. Now you have art which is the foundation of who humans are Yeah, so that's what art is art is life because art is all three art is a way to minimize surprises That's why beautiful music. We know what it is because it is literally an expression of life I think eventually we'll do a podcast on that subject at some point because I that's something I've been very interested in as Of late as I was trying just learn in a vacuum without risk. We're out. Yes We try to do all our learning when the stakes are higher like training. That's a tough game It's good to learn how to learn but that's very important because then we can go into anecdotal versus logical versus empirical evidence because if you look people now The trainers all try to look to learn through empirical evidence, which means they want to read something that makes them change the prediction But without having to make the world fit, so without having to do anything with it, oh how they feel about it So it's a laziest way to learn. You can't learn just through empirical evidence your own shit He's saying it's thinking that you're going to do scrambled eggs do Gordon Ramsay way because you read the learning how other people do things Yeah, and you but you still haven't learned how he did it Just cause you read about Godzilla runs his life. And you read his recipe does I mean you can do the scrambled eggs the way he did it That's the problem is that's what they want is like oh, but just tell me how you did it It doesn't matter because that's not how you're going to do it because that's not how this works So you think changing the position gives you control of life where you are. So mistaken, it is only one When is the dysfunction anytime you don't use all three you are being dysfunctional. So whenever you think empirical evidence will give you truth What were you doing basically is to be morally superior to everybody else your new ivory tower Where you think you can bring the truth to everybody like your Moses on Mount Sinai basically because you want an easy way to mitigate surprises Which are actually is the other two that you need to go to put yourself through you'll see that most dysfunction results in people relying on one way to change To mitigate surprises either they change your prediction which means either they quit on everything or they want everything without actually having to work Or they just use movement but then keep the same behavior going that fucked up behavior But just keep

acting on it trying to fix it without Even realizing where they fuck up in the first place Right, or they basically have this function of trying to change the way they feel drugs food Mental illness anxiety all that stuff. That's what this function is is when you do one optional over the others I Think we just summed up all of Carl Fuerst ins working like an hour and 20 minutes Well, you wish but yeah, no I I hope so by all I mean the two paragraphs of his stuff that I had to reference today Yeah, but that's that's where we we talk about about it in that online the nervous system Yes workshop that we're going to release in a in a week or two But this time we get out we should be able we should really sit This is the base of everything. I do like but now I have the neurological understanding of it So this is where strong fit is going is going into behavior modification. It's going to work hard to be Alive I guess but the progress I've made on myself because I basically I keep getting leaner because I apply basically that to nutrition and it The effects are ridiculous. So now I'm playing pull so much better because I'm applying that when I play pool. I Understand where my dysfunctions were before. So what that's what OCD is OCD is option to make the world fit right? So you're like no No, I need my hand here mine here because I think the only way to mitigate surprises is to do something physical about it That's OCD for you Give me what I'm not sure whether or not I Started the cameras and I have no but no matter while I went up and looked at right in the face. Yeah, yeah Because you know why because you don't want to surprise yeah fucking up Yeah, right But so the you need a little bit of option number three, which would be like dude relax You know you it I've only done it once it's only one cancer. Exactly and I think anyone even complained on YouTube So but you need obviously you need to change your prediction in order to not make the same mistake again, right? So as you see it's that it's so sweet constantly in balance. That's where you make progress So that's I've been applying that two points ridiculous like this morning. I had a really good session based on that. Oh, Yeah, it was well So now the fun stuff. Yeah. All right, so that's gonna have us wrapped up for this week. Yeah, I think so You guys follow strong fit one on Instagram and you follow me at Tyler effing stone on Instagram Strong fit equipment calm that's where we have all the sandbags. Payroll t-shirts t-shirt. Come on you won't either so we got a We have more cool t-shirts cutting. So yeah, so we've got but also now we do have strong fit equipment EU. Yeah, so for Europe You want to make some limited apparel there? We'll get more stuff in there good time But but that's the stuff that's coming directly out of this office here. So, okay, so it's shipping is better. Yes. Yes, so We have the strong feet community and that's very important Anyway, I heard a lot of people saying they heard about it to the podcast. So we have to imagine it good That's my

that's actually probably my favorite form we have we have some some private stuff here and there but but that group is nice because that's people who have Seen some stuff and are excited enough to get in we're not yelling to to a roomful of people who don't give a shit There's a lot of like-minded people. Yes, really good We're gonna release those so the autoregulation Nutrition group number two is starting on April 1st, and then we are just so we can tell people we're starting in the autoregulation Training group. Yes on May 3. So in the next podcast, I'll explain what that is. Yeah let's see where the training group nervous system work the online nervous system work shot is that we're gonna release that you know week or Two as well. Ok I think we talked so Ben your neck podcast we can we can start by talking about the training Perfect, because that kind of fits because we need to talk about intent. Yeah, I like it when it comes to training Yeah, so that's next week next week. We mean like tomorrow for us Yeah tomorrow, but that's what I have a different t-shirt and I'll fake it like it's next week of shower. I promise. All right everybody Good you need something to drink